UNIVERSITY OF MADRAS

DEPARTMENT OF COMPUTER SCIENCE,

Common to MCA , MSc(CS) and MSc(IT) .

End Semester Examination – December -2024

Sub Code: MSI E-357 , Software Project Management and Testing

Date: 13-12-2024 , Total Marks=80, Time: 3 hrs.

Answer Any EIGHT Questions , Mark : 08 X 10 = 80

1. Explain what is software project and explain different SDLC (software development life cycle) in software project management . (10 mark)

2. Explain what is project planning and explain step wise different project planning in software development. (10 mark)

3. Explain (i) Digital and making india movement (5 + 5 mark)
   (ii) Draw and explain any Banking system in E-R diagram

4. Explain (i) what are the benefits involved in software project
   (ii) Draw and Explain University Administration Project (5 + 5 mark)

5. Explain (a) White Box Testing techniquies (5 mark)
   (b) Structural Testing techniquies (5 mark)

6 Explain (a) Unit Testing techniquies (5 mark)
   (b) Integration Testing techniquies (5 mark)

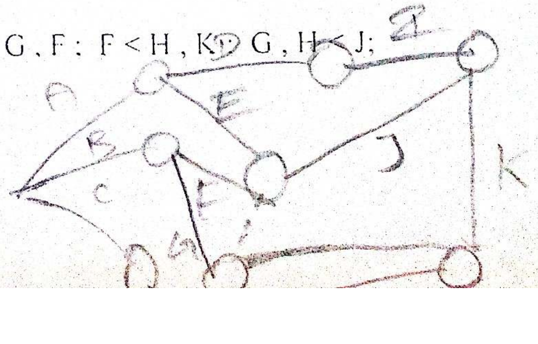
7. Explain (a) System Testing techniquies (5 mark)
   (b) Debugging techniquies (5 mark)

8. Construct the project network of the activities given below (5 + 5 mark)

(a).

Activity :	A	B	C	D	E	F	G	H	I	J	K
Immediate Predecessor :	-	-	-	A	B	B	C	D	E	H,I	F,G

(b) A < C . D . I ; B < G . F ; D < G . F ; F < H , K; G , H < J;
I . J . K < E



09) Calculate the earliest start, earliest finish, latest start and latest finish of each activity of the project given below and determine the Critical path of the project. (10 mark)

Activity	1-2	1-3	1-5	2-3	2-4
Duration (in weeks)	8	7	12	4	10

Activity	3-4	3-5	3-6	4-6	5-6
Duration (in weeks)	3	5	10	7	4

10) Construct the network for the project whose activities and the three time estimates of these activities (in weeks) are given below, Compute (10 mark)

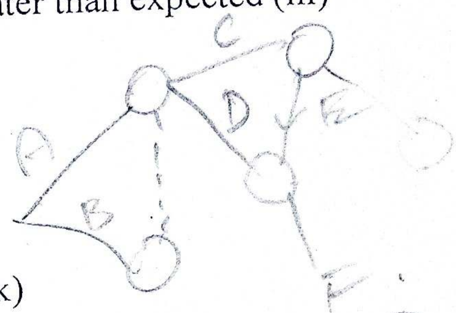
(a) Expected duration of each activity
(b) Expected variance of each activity
(c) Expected variance of the project length

Activity	t_o	t_m	t_p
1-2	3	4	5
2-3	1	2	3
2-4	2	3	4
3-5	3	4	5
4-5	1	3	5
4-6	3	5	7
5-7	4	5	6
6-7	6	7	8
7-8	2	4	6
7-9	1	2	3
8-10	4	6	8
9-10	3	5	7

11) Three timeestimates (in months) of all acitivites of a project are as given below: (10 mark)

Activity	Time in Months a	m	b
1-2	0.8	1.0	1.2
2-3	3.7	5.6	9.9
2-4	6.2	6.6	15.4
3-4	2.1	2.7	5.1
4-5	0.8	3.4	3.6
5-6	0.9	1.0	1.1

(i) Construct the project network and find the probability that the project will be completed in (ii)Two month later than expected (iii) not more than 3 month earlier than expected.

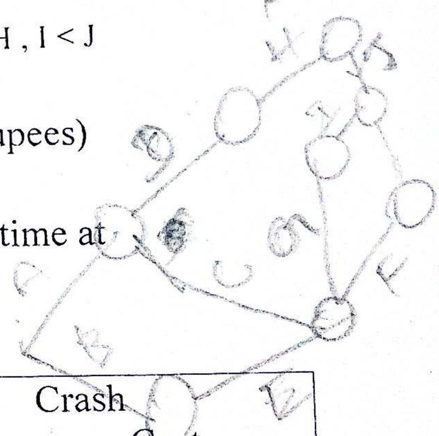


12) (a) Construct the project network (5 mark)

A < C, D ; B < C, D ; C < E ; D , E < F

(b) Construct the project network (5 mar k)

A < C , D ; B < E; C , E < F , G ; D < H ; G < I ; H , I < J



13) The following time-cost table(time in days, cost in rupees) applies to a project. Use it to arrive at the network associated with completing the project in minimum time at minimum cost. (10 mark)

Activity	Normal Time	Normal Cost	Crash Time	Crash Cost
1-2	2	800	1	1400
1-3	5	1000	2	2000
1-4	5	1000	3	1800
2-4	1	500	1	500
2-5	5	1500	3	2100
3-4	4	2000	3	3000
3-5	6	1200	4	1600
4-5	3	900	2	1600